

Mobile Device Security Models

An in-depth analysis of Android, iOS, and Windows security architectures and enterprise management trends.

COMMON SECURITY PILLARS

Access Control

Strictly controlling access to applications and downloads. Vendors focus on application provenance to ensure that software comes from trusted, verified sources before reaching the device.

Compartmentalization

Isolating applications and their resources once downloaded. This ensures that even if one application is compromised, the rest of the system and sensitive user data remain protected.

The three major vendors—Google, Apple, and Microsoft—share these core philosophical foundations.

Google Android Security Architecture

Built on the Linux Open Source Foundation

THE ANDROID SANDBOX

Kernel-Level Isolation

Android adapts the Linux multiuser environment for application protection:

- Each app is assigned a **Unique User ID (UID)**.
- Apps run as separate user processes in their own **Dalvik VM**.
- The Linux kernel prevents "User A" from reading "User B's" files.
- **Process Sandboxing**: Ensures a memory crash in one app does not compromise the device.

PERMISSIONS & DEVICE INTEGRITY



SDK Security

Features like **ProPolice** prevent stack overflows. Hardware-based "No-eXecute" parameters stop code execution on the stack or heap.

Software Development Kits (SDK)



Permission Model

Apps must request access to protected APIs (Camera, GPS, SMS) in their manifest. Prior to Android 6.0, permissions were granted permanently at installation.



Rooting Risks

Rooting grants unrestricted access to the kernel. This circumvents the sandbox and significantly increases vulnerability to malware and rootkits.

THE EVOLUTION OF ENCRYPTION

A large, bold, light blue percentage sign graphic, specifically '80%', is displayed on the left side of the slide. It is positioned above the text 'Devices Secured in Nougat'.

Devices Secured in Nougat

Mandatory Encryption

Early Android versions had low encryption rates (approx. 25%). Google's insistence on encryption-by-default in Android 7.0 (Nougat) dramatically improved security.

For low-end IoT devices and cheap smartphones, Google introduced **Adiantum**—a highly efficient encryption algorithm designed for resource-constrained hardware.

Apple iOS Security Model

The "Walled Garden" Strategy

APPLICATION PROVENANCE

Integrity Through Control

Apple uses digital certificates to sign applications. Only authorized developers can upload to the App Store, where apps undergo rigorous testing.

This approach deters malicious code by holding developers accountable and ensuring that release code cannot be tampered with after approval.

*Model only applies to non-jailbroken devices.

IS ANTI-CONSUMER: HERE'S WHY



IOS SANDBOXING & PRIVACY

-  **Chrooted Access:** Unlike Android's UID model, all iOS apps run as the same user ("mobile") but are restricted to their own specific filesystem slice.
-  **Layered Encryption:** Uses a layered approach where sensitive personal data (emails/messages) is double-encrypted to prevent physical access leaks.
-  **Granular Permissions (iOS 13+):** Introduced choices like "Allow Once" and "Only while using the app" to mitigate background tracking concerns.
-  **Sandbox Isolation:** Apps cannot even determine if other applications are present on the device, providing superior metadata protection.

COMPARATIVE ANALYSIS

Security Feature	Google Android	Apple iOS	Windows Phone (8.1)
OS Foundation	Linux Kernel	UNIX (FreeBSD)	Windows NT Kernel
App Isolation	UID-based Sandbox	Chrooted Sandbox	Process-based Isolation
App Source	Open (Any site)	Walled Garden (Store)	Closed (Store)
Permission Request	Installation time (Legacy)	Execution time	Varies (Store-vetted)
Rooting Term	Rooting	Jailbreaking	Unlocking

ANDROID SECURITY EVOLUTION



THANK YOU

